Corporate Finance

Lecture 4: Time-Varying Rates of Return

Beginner-friendly study notes (from CF_4)

How to use these notes

1. Read in order. Each part starts with the plain-English idea, then the formula, then an easy example.
2. Try the **Check yourself** questions at the end of each part. Answers are at the back (Part 9).
3. This lecture continues the earlier one (PV, NPV, IRR, forward and spot rates). Where it helps, a quick reminder is given.
4. "Units" or "rupees" just means money.

Contents

Part	Topic
1	Maturity and duration: how long does a project really last?
2	Inflation and how it is measured: CPI, nominal vs real GDP, GDP deflator, WPI, core inflation
3	Nominal vs real interest rates, and inflation inside NPV
4	The yield curve: spot rates, holding vs annualized returns, forward rates
5	Investment horizon: what if you sell a bond early?
6	Interest rate risk: why long bonds are riskier
7	Why yield curves slope: term premium, shapes, expectations, TIPS
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Part 1. Maturity and duration

Big idea
We want **one number** that says how long a project (or bond) lasts. A bond that pays 1 rupee in year 1 and 1 rupee in year 30 is clearly very different from one that pays 100 in year 1 and 1 in year 30, yet both end in year 30.

1.1 Maturity: the simplest measure

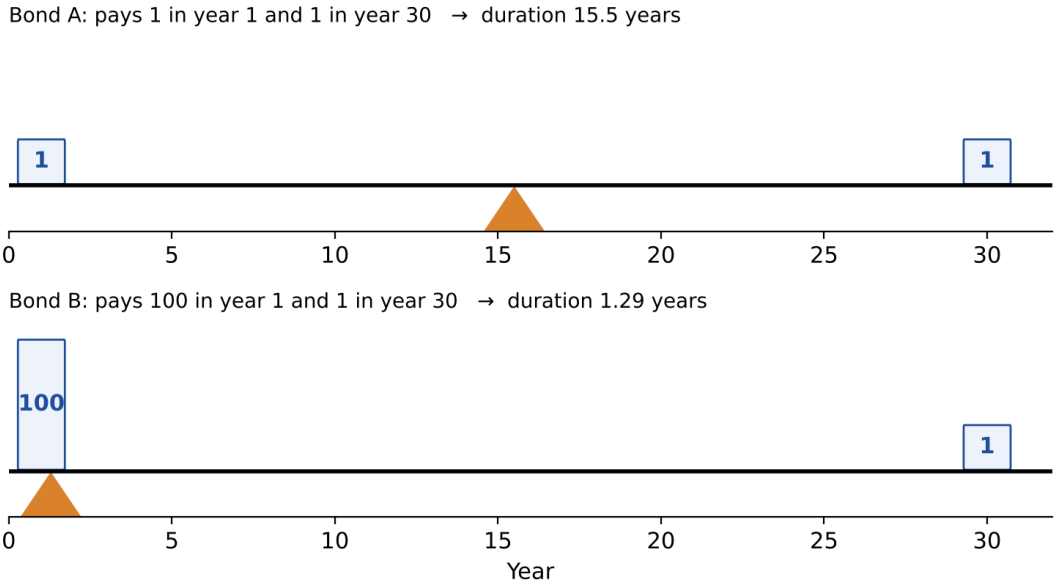
Maturity = the date of the *last* payment. All three bonds in the lecture (1 in year 30; 1 in year 1 and 1 in year 30; 100 in year 1 and 1 in year 30) have a maturity of 30 years. So maturity cannot tell them apart. That is its weakness.

1.2 Duration: the payment-weighted average time

Formula
Duration = $\Sigma\ (C_t \times t) \div \Sigma\ C_t$ (C_t = payment at year t)

Think of a seesaw. Each payment is a weight placed at its payment date. Duration is the **balance point**.

Duration = the balance point of the payments on a time line



The big payment in year 1 pulls the balance point very close to year 1.

Worked example: From the lecture
Bond A: 1 in year 1, 1 in year 30. Duration = (1×1 + 1×30) ÷ (1+1) = **15.5 years**.
Bond B: 100 in year 1, 1 in year 30. Duration = (100×1 + 1×30) ÷ 101 = **1.29 years**.

1.3 Macaulay duration: use present values

Plain duration treats a rupee in year 30 the same as a rupee in year 1. But we know later money is worth less. **Macaulay duration** fixes this by weighting with **present values** instead of raw payments.

Formula
Macaulay Duration = Σ (PV(C_t) × t) ÷ Σ PV(C_t)

- It depends on interest rates, so it **changes when rates change**.
- Far-away payments get discounted heavily, so they count for less in the average.

Worked example: From the lecture
Pays 100 next year and 1 in 30 years. 1-year rate 2%, 30-year rate 5%.
PV(100) = 100 ÷ 1.02 = 98.04. PV(1) = 1 ÷ 1.05^{30} = 0.23. Total value = 98.27.
Macaulay duration = (98.04×1 + 0.23×30) ÷ 98.27 ≈ **1.07 years**.

Check yourself – Part 1

- Q1. Find the duration of a bond paying 50 in year 1 and 50 in year 10, and one paying 90 in year 1 and 10 in year 5.
- Q2. A project pays 100 in year 1 and 100 in year 2. Spot rates: 1-year 4%, 2-year 5%. Find the Macaulay duration.

Part 2. Inflation and how it is measured

Big idea
Inflation = a sustained rise in the general level of prices. When prices rise, each rupee buys less. If inflation is 10% for all goods, something costing 100 today costs 110 next year. The **inflation rate** is how fast the price level rises.

2.1 Consumer Price Index (CPI): cost of living

CPI tracks the cost of a "typical household basket" of goods and services. Method: (1) survey households to decide what is in the basket; (2) collect prices of every item in the base year and the current year; (3) compare costs.

Formula
CPI = (cost of basket in current year ÷ cost of basket in base year) × 100 The **quantities stay at base-year levels**.

Worked example: A toy basket
Basket: 20 kg rice, 10 litres milk, 5 litres fuel. Base prices: 50, 40, 80. Current prices: 60, 44, 96.
Base cost = 1000 + 400 + 400 = 1800. Current cost of the *same basket* = 1200 + 440 + 480 = 2120.
CPI = 2120 ÷ 1800 × 100 = **117.78**. Living costs rose by 17.78%.

CPI as a weighted average of price changes

The lecture shows that CPI/100 = $\Sigma f_i \times (p_C^i \div p_B^i)$, where f_i is the share of good i in the base-year basket. In the toy basket the weights are rice 1000/1800 = 55.6%, milk 22.2%, fuel 22.2%, and the price ratios are 1.20, 1.10, 1.20: 0.556×1.20 + 0.222×1.10 + 0.222×1.20 = 1.178 (same answer). **Goods that take a bigger share of spending matter more.**

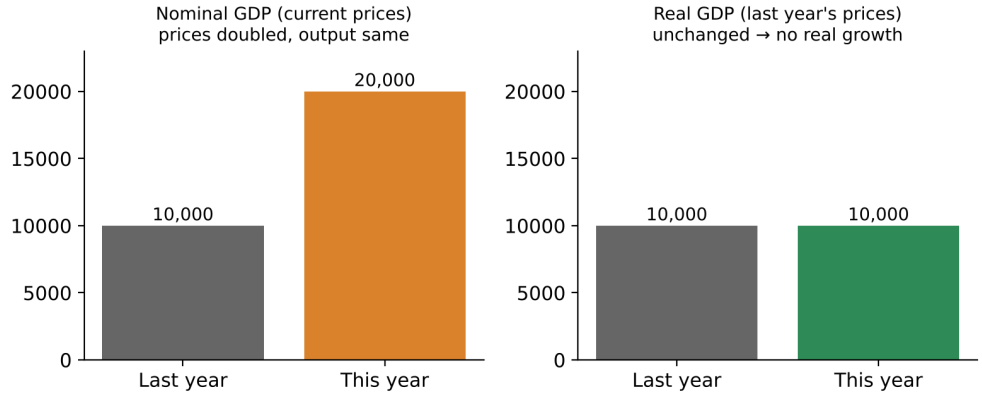
The lecture's table gives India's CPI weights (2012): food & beverages is 36.29% for urban households but 54.18% for rural; housing is 21.67% (urban); fuel and light 5.58% / 7.94%. Rural households therefore feel food-price changes more.

Limits of CPI
CPI measures the cost of a **fixed** basket, which may not equal the cost of maintaining a standard of living. Three issues: (1) **substitute bias** (people switch away from goods that become expensive, but the basket does not), (2) **new goods** appear, (3) **unmeasured quality changes**.

2.2 Nominal GDP and real GDP

GDP = market value of all final goods and services produced in a period = Σ price × quantity. **Nominal GDP** uses current prices. **Real GDP** uses prices of a fixed *base year*, so it changes only when quantities change.

Worked example: Chocolates and shoes (lecture)
Last year: 50 chocolates at 100 and 20 shoes at 250. Nominal GDP = 5000 + 5000 = **10,000**.
This year: same output (50 and 20), prices now 200 and 500. Nominal GDP = 10,000 + 10,000 = **20,000**. It doubled, but nobody produced more!
Real GDP this year (last year's prices) = 50×100 + 20×250 = **10,000**. No real growth.
If instead output rose to 100 chocolates and 40 shoes, real GDP = 100×100 + 40×250 = **20,000**: real growth of 100%.



Nominal GDP mixes price and quantity changes; real GDP isolates quantity.

2.3 GDP deflator

Formula
GDP deflator P_t = Nominal GDP in year t ÷ Real GDP in year t

If nominal GDP rises faster than real GDP, the extra comes from higher prices. In the example above the deflator is 20,000 ÷ 10,000 = **2.0**: prices doubled.

GDP deflator vs CPI. They are not the same: the deflator covers all final goods produced in the economy (including things sold not to consumers, e.g. machines), while consumers buy a particular basket that also includes imported goods.

2.4 Other price measures

Measure	What it is
WPI (Wholesale Price Index)	Prices paid to wholesalers by other producers. India's basket: primary articles 22.62, fuel and power 13.15, manufactured products 64.23 (weights).
Core inflation	Long-run trend in prices, excluding volatile items like food and fuel. In India this removes a large part of the basket.

Check yourself – Part 2

- Q3. Using the toy basket idea: base quantities 20 rice, 10 milk, 5 fuel; base prices 50, 40, 80; new prices 60, 44, 96. Find the CPI, the inflation rate and rice's weight.
- Q4. Base-year prices: laptop 20, car 50. Last year: 30 laptops, 10 cars. This year prices are 25 and 60, output is 40 laptops and 12 cars. Find nominal GDP both years, this year's real GDP (base = last year's prices), the deflator and real growth.

Part 3. Nominal vs real interest rates

Big idea
When you borrow or lend, what really matters is **how many goods** you give up or gain, not how many rupees. Inflation makes these different. A 10% interest rate with 10% inflation means you gain nothing in goods.

Type	Measured in	Note
Nominal rate i_t	Currency. Borrow 1 rupee, repay $1 + i_t$ next year.	The rate in newspapers and bank ads
Real rate r_t	Baskets of goods. Borrow 1 basket, repay $1 + r_t$ baskets next year.	The rate that matters for saving, investing and macroeconomics

3.1 The link between them (Fisher relation)

Suppose a good costs P_t today. To buy one unit now you borrow P_t and repay $P_t(1+i_t)$ next year. Next year the good is expected to cost P^e_{t+1}, so the repayment in goods is $P_t(1+i_t) \div P^e_{t+1}$. That is $1 + r_t$.

Formula
$1 + r = (1 + i) \div (1 + \pi^e)$ Approximation (small rates): $r \approx i - \pi^e$ (π^e = expected inflation)

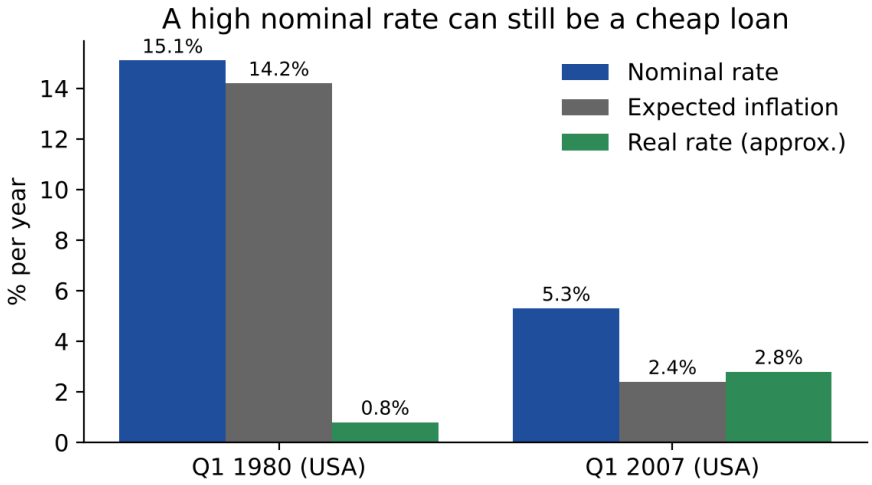
Worked example: Easy numbers
Nominal 8%, expected inflation 5%. Exact real rate = $1.08 \div 1.05 - 1$ = **2.86%**. Approximation: $8 - 5 = 3\%$. Close.

Worked example: Lecture: when the approximation fails
Inflation 100%, nominal bond rate 700%. One rupee becomes 8 rupees. But prices double, so you need 2 rupees per basket: 8 rupees = 4 baskets.
Real rate = $(4 - 1) \div 1$ = **300%**. Exact formula: $8 \div 2 - 1 = 300\%$. The shortcut $700 - 100 = 600\%$ would be badly wrong. **Use the exact formula when rates are large.**

How do we get expected inflation?

- **Static**: inflation will not change.
- **Adaptive**: people extrapolate from recent experience.
- **Rational**: expectations consistent with the model of the economy.
- **Data-based**: from surveys or financial-market data.

Worked example: Was borrowing cheaper in 1980 or 2007? (lecture)
USA Q1 1980: nominal 15.1%, expected inflation 14.2% → real rate ≈ **0.8%**. Q1 2007: nominal 5.3%, expected inflation 2.4% → real ≈ **2.8%**.
So in real terms, **borrowing was cheaper in 1980**, even though the headline rate was three times higher.



Headline (nominal) rates mislead; compare real rates.

3.2 Inflation inside NPV: the golden rule

Match the rate to the cash flows
Nominal cash flows → discount with the nominal rate.
Real cash flows → discount with the real rate.
Never mix them (nominal flows with a real rate, or the reverse).

Worked example: Lecture example 1B
Nominal rate 700%, real rate 300%, inflation 100%. A project gives 12 baskets next year (= 24 rupees, since a basket will cost 2 rupees).
Method 1 (real flows, real rate): 12 ÷ (1 + 300%) = **3 baskets** today.
Method 2 (nominal flows, nominal rate): 24 ÷ (1 + 700%) = **3 rupees** today (and a basket costs 1 rupee today, so the same value).

Worked example: My own example
A project pays 1,100 rupees next year. Nominal rate 10%, inflation 4%. Real rate = 1.10 ÷ 1.04 − 1 = 5.77%.
Nominal way: 1,100 ÷ 1.10 = **1,000**. Real way: real cash flow 1,100 ÷ 1.04 = 1,057.69, discounted at 5.77%: 1,057.69 ÷ 1.0577 = **1,000**. Same.
Mixing (1,100 at the real 5.77%) gives 1,040: **wrong**.

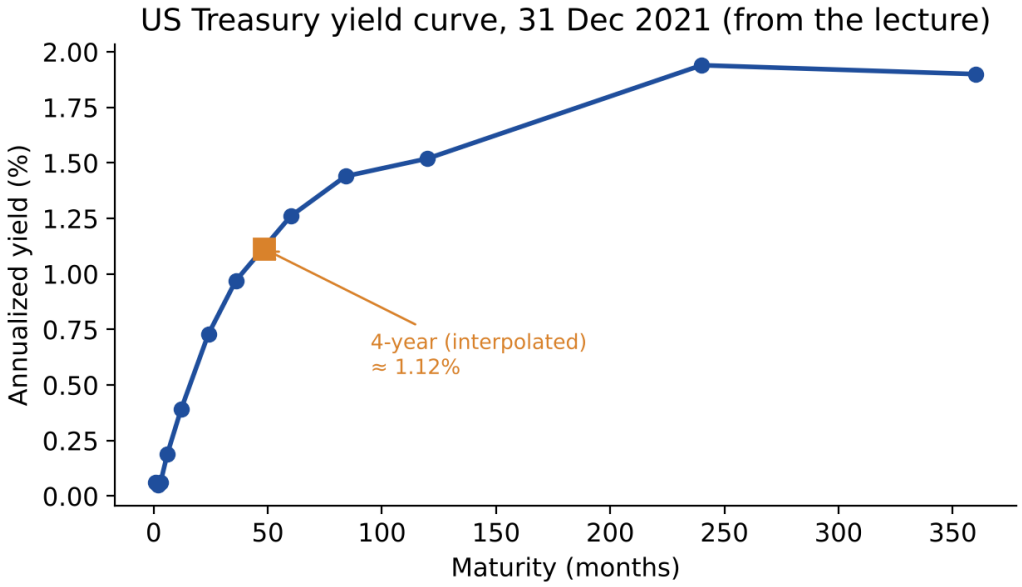
Check yourself – Part 3

- Q5. Nominal rate 12%, expected inflation 5%. Find the real rate exactly and by approximation.
- Q6. A bond pays a nominal 300% when inflation is 100%. What is the real rate?
- Q7. A project pays 1,100 next year. Nominal rate 10%, inflation 4%. Show that discounting nominal flows with the nominal rate equals discounting real flows with the real rate.

Part 4. The yield curve

Big idea
A 1-month, a 5-year and a 30-year government bond usually pay **different** annual rates. The **yield curve** (also called the **term structure of interest rates**) plots the annualized rate against the time to maturity.

The lecture assumes **zero bonds**: bonds with no coupon payments in between. You pay a price now and get one payment at maturity.



US Treasury curve from the lecture. Rates rise with maturity (upward-sloping).

Maturity	1m	2m	3m	6m	1y	2y	3y	5y	7y	10y	20y	30y
Yield %	0.06	0.05	0.06	0.19	0.39	0.73	0.97	1.26	1.44	1.52	1.94	1.90

4.1 Three kinds of return (do not mix them up)

Name	Symbol	Meaning
Holding return	$r_{0,t}$	Total return from today to year t
Annualized return	r_t	Single yearly rate that gives the same total. What the yield curve shows
Individual annual return	$r_{t-1,t}$	Return earned in one particular year

Formula
$1 + r_{0,t} = (1 + r_t)^t$

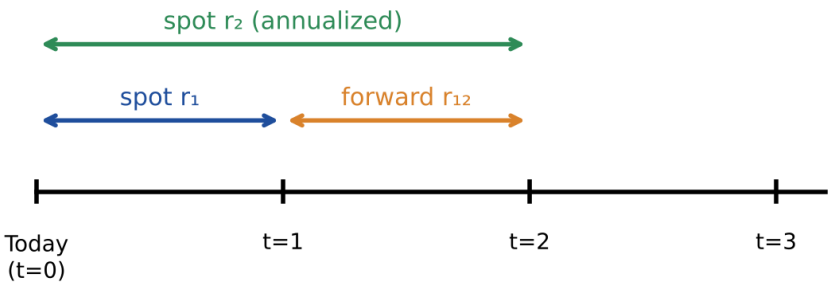
Worked example: Lecture examples
30-year bond (1.90%): 100 becomes $100 \times (1.019)^{30}$ = **175.88**. Holding return = 75.9%.
3-month bond (0.06%): $100 \times (1.0006)^{1/4}$ = 100.015. (Only a quarter of a year, so only a quarter-power.)
2-year bond (0.73%): $r_{0,2} = (1.0073)^2 - 1$ = 1.465%. So 100,000 becomes **101,465**.

Interpolation: yields at maturities not listed

There is no 4-year point, so estimate it as the average of 3-year (0.97%) and 5-year (1.26%): (0.97 + 1.26) ÷ 2 ≈ **1.12%**. Only approximate, because the curve is not a straight line.

4.2 Forward rates: the rate that starts in the future

Spot rates start today; forward rates start in the future



$$(1 + r_2)^2 = (1 + r_1)(1 + r_{12})$$

A **forward rate** $r_{1,2}$ is the rate for the year starting at t=1 and ending at t=2. Today's prices of 1- and 2-year bonds already imply it:

Formula

$(1 + r_2)^2 = (1 + r_1)(1 + r_{1,2}) \quad \Rightarrow \quad 1 + r_{1,2} = (1 + r_2)^2 \div (1 + r_1)$

Worked example: Lecture: from the Dec 2021 curve

$r_1 = 0.39\%$, $r_2 = 0.73\%$. Forward = $(1.0073)^2 \div 1.0039 - 1 =$ **1.07%**.
In the same way $(1.0097)^3 = (1.0039)(1.0107)(1 + r_{2,3})$ gives $r_{2,3} \approx 1.46\%$.

Worked example: Easy numbers

1-year rate 5%, 2-year rate 10%. Forward = $1.10^2 \div 1.05 - 1 = 1.21 \div 1.05 - 1 =$ **15.24%**. (You earn 5% in year one, then must earn 15.24% in year two to match 10% a year on the 2-year bond.)

Check yourself – Part 4

- Q8. Using the 3-year yield of 0.97%, how much does 50,000 become after 3 years? What is the holding return?
- Q9. Estimate a 15-year yield by interpolating between 10-year (1.52%) and 20-year (1.94%).
- Q10. 1-year rate 3%, 2-year 4%, 3-year 5%. Find the forward rates $r_{1,2}$ and $r_{2,3}$.

Part 5. Investment horizon and bond payoffs

Question: You need your money back in 1 year. A 1-year zero bond yields 0.39%; a 3-year zero bond yields 0.97% a year. Can you buy the 3-year bond anyway?

Yes: you can sell it after one year. But in a perfect and certain market you do **not** earn more than 0.39%. Why? Because the price you sell at next year already reflects the forward rates.

Worked example: Lecture numbers (bond pays 100)
Price today of the 3-year bond: $100 \div (1.0097)^3 = \mathbf{97.15}$.
Next year it is a 2-year bond with forward rates 1.07% and 1.46%: price = $100 \div (1.0107 \times 1.0146) = \mathbf{97.52}$.
1-year holding return = $(97.52 - 97.15) \div 97.15 \approx \mathbf{0.39\%}$: exactly the 1-year bond's return.

Worked example: Easy version
Rates: 1-year 3%, 2-year 4%. Buy a 2-year zero for $100 \div 1.04^2 = 92.46$. After one year the forward rate (5.01%) prices it at $100 \div 1.0501 = 95.23$. Return = $95.23 \div 92.46 - 1 = \mathbf{3.00\%}$, same as the 1-year bond.

Big idea
In a world with certainty, **every bond gives the same return over any single period**. Differences in yields only reflect when the money is paid. In the real world, future rates are uncertain, which brings risk (Part 6).

Check yourself – Part 5

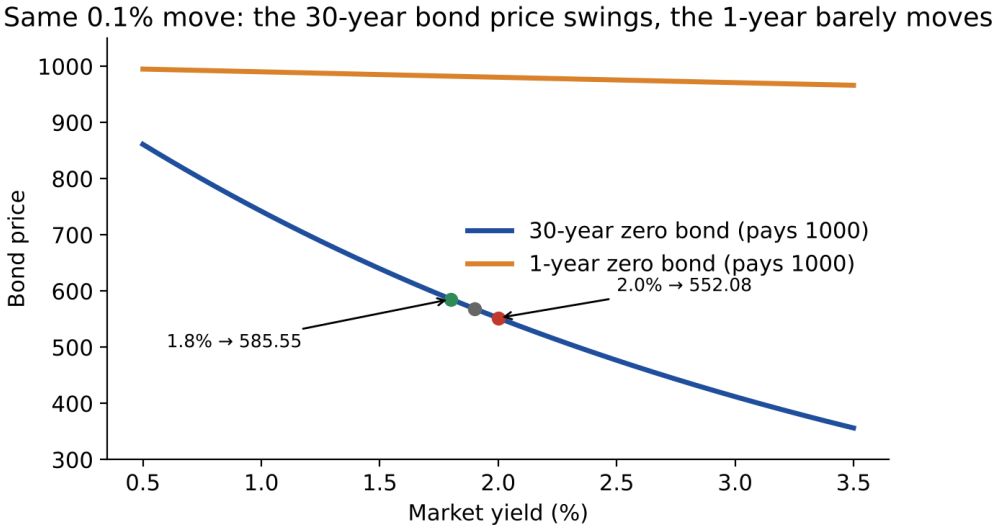
- Q11. 1-year rate 3%, 2-year rate 4%. A 2-year zero pays 100. Find its price today, its price after one year (using the forward rate), and the 1-year return.

Part 6. Interest rate risk: why long bonds are riskier

Government bonds have no default risk, but they have another risk: **interest rates move**. If the market rate rises, existing bonds (paying the old, lower rate) fall in price, and vice versa. **Equal-sized rate changes hit long-term bonds much harder.**

Basis point (bp) = 0.01%. So 10 basis points = 0.10%.

	30-year zero (pays 1000)	1-year zero (pays 1000)
Price at 1.90%	$1000 \div 1.019^{30} = \mathbf{568.56}$	$1000 \div 1.019 = \mathbf{981.35}$
Yield rises to 2.0%	552.08 → **−2.9%**	980.39 → **−0.1%**
Yield falls to 1.8%	585.55 → **+3.0%**	982.32 → **+0.1%**
Risk per 10 lakh invested	about 30,000 per 10bp move	about 1,000 per 10bp move



Price vs yield. The 30-year bond's price is steep and curved; the 1-year bond's is nearly flat.

- A **short-term bond** returns your money soon, and you can reinvest at the new rate. Little risk.
- A **long-term bond** can gain or lose a lot of value. Only the shortest-term bond is truly risk-free.
- This is why duration matters: longer duration means more sensitivity to rate changes.

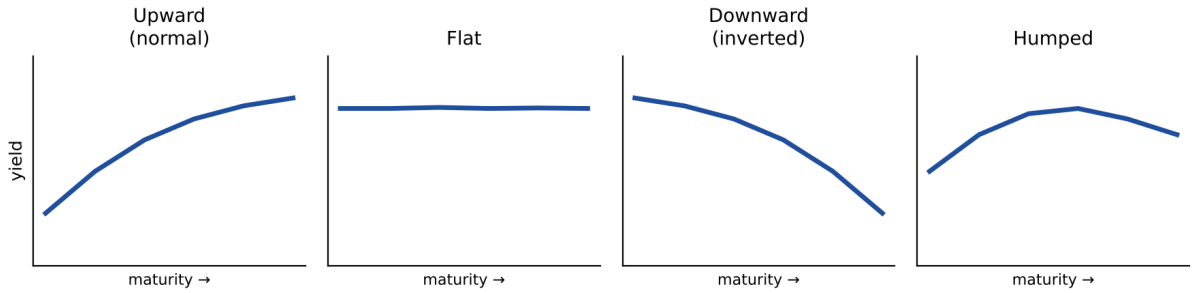
Check yourself – Part 6

- Q12. A 20-year zero pays 1000. At 2% find its price. Then find the % price change if the yield rises 10bp, and if it falls 10bp. Compare with a 1-year zero for the same moves.

Part 7. Why do yield curves slope? Term premium and shapes

Term-varying cost of capital: the right discount rate depends on how long the project is. The curve is usually **upward-sloping**: long projects face a higher cost of capital even without extra default risk. The extra compensation is the **term premium** (or time premium) = long-term rate minus short-term rate.

7.1 Four shapes



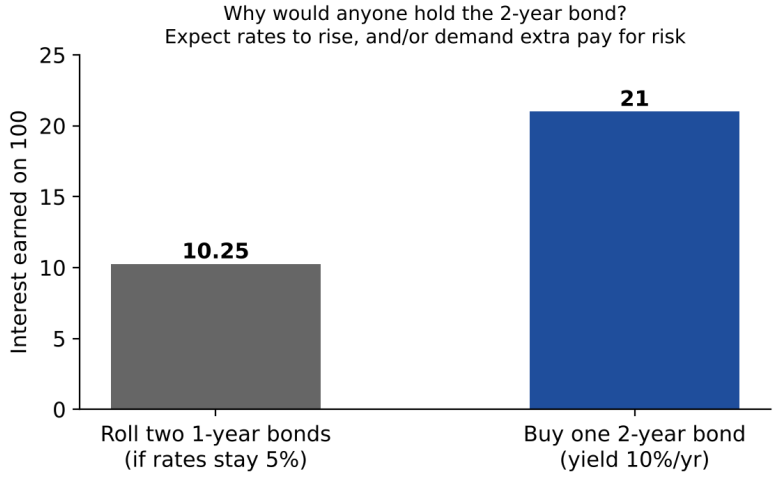
Shape	Meaning	Lecture example
Upward (normal)	Short rates below long rates. Most common. Steep curves often follow a recession.	May 2011
Flat	Little difference between short and long rates	January 2007
Downward (inverted)	Short rates above long rates. Sometimes signals an upcoming recession.	December 1980
Humped	Short and long rates above medium-term rates	June 1979

The yield curve is a useful but **noisy and unreliable** signal of where the economy is heading. It is set by supply and demand for savings and credit; central banks influence but do not control it.

7.2 Two explanations for an upward slope

Take 1-year rate 5% and 2-year rate 10%. The forward rate is 15.24%. Why would investors accept this?

Explanation	Idea	In equation form
1. Expectations	The 1-year rate next year really will be about 15.24%. Long rates are just a forecast of higher future short rates.	$(1 + r_{1,2})(1 + E(r_{2,3})) = (1 + r_{1,3})$
2. Risk premium	Investors demand extra return to hold risky long bonds.	$(1 + r_{1,2})(1 + E(r_{2,3})) < (1 + r_{1,3})$



Rolling two 1-year bonds at 5% earns 10.25 on 100; the 2-year bond earns 21.

Notation as in the slides. If the curve keeps its shape, two rolled 1-year bonds earn 10.25, while the 2-year bond earns 21. The gap must be explained by higher expected future rates and/or compensation for risk.

7.3 Is it just inflation? TIPS

- Investors demand higher nominal rates when they expect higher inflation, which could make nominal curves slope upward.
- But **inflation-indexed bonds** (e.g. TIPS in the US) pay the promised rate *plus* CPI inflation. They are not hurt by inflation, so if expected inflation were the only reason, their real yield curve would be **flat**.
- End-2021 TIPS yields: 5-year −1.61%, 7-year −1.31%, 10-year −1.04%, 20-year −0.63%, 30-year −0.44%. The real curve was **even steeper** than the nominal one, so inflation expectations cannot be the whole story (in most of history they cannot, though sometimes they can).

7.4 What does the evidence say?

- A high forward rate does **not** reliably predict a high future interest rate.
- Historically, buying 30-year bonds beat rolling one-month bonds for 30 years. Why?
- **Segmented markets** (different investors like different bonds) and **irrational investors / taxes / liquidity**: judged unlikely.
- **Most likely: compensation for risk.** Long bonds swing more in price (Part 6), while rolling short bonds protects you from rate changes (if rates double, you reinvest at higher rates). So long bonds must pay more on average.

Check yourself – Part 7

- Q13. Name the shape: (a) 1-year 5%, 10-year 3%. (b) 1-year 2%, 5-year 3%, 30-year 2%. (c) 1-year 1%, 30-year 4%.
- Q14. 1-year rate 5%, 2-year rate 10%. Suppose the market expects next year's 1-year rate to be 8%. How much extra (over two years) does the 2-year bond pay versus rolling 1-year bonds? What does this extra represent?

Part 8. Formula cheat sheet

Topic	Formula	Remember
Duration	$\Sigma C_t t \div \Sigma C_t$	Balance point of payments
Macaulay duration	$\Sigma PV(C_t)t \div \Sigma PV(C_t)$	Uses present values; depends on rates
CPI	(basket cost now ÷ basket cost base) × 100	Fixed base quantities
CPI as weights	$\Sigma f_i (p_C^i / p_B^i)$	f = base spending share
Nominal GDP	$\Sigma p_{current} q_{current}$	Prices and quantities change
Real GDP	$\Sigma p_{base} q_{current}$	Only quantities change
GDP deflator	Nominal GDP ÷ Real GDP	Price level of output
Real interest rate	$1 + r = (1 + i) \div (1 + \pi^e)$ $r \approx i - \pi^e$	Use exact when rates are big
NPV with inflation	Nominal flows / nominal rate; real flows / real rate	Never mix
Holding vs annualized	$1 + r_{0,t} = (1 + r_t)^t$	Annualized = t-th root
Forward rate	$1 + r_{1,2} = (1 + r_2)^2 \div (1 + r_1)$	Rate that starts in the future
Zero bond price	$P = Face \div (1 + r_t)^t$	Long maturity: price very sensitive to r
Term premium	long-term rate − short-term rate	Extra pay for long-term risk

Part 9. Practice problems

Try each one before looking at the answers.

Q1. Duration: (a) a bond pays 50 in year 1 and 50 in year 10; (b) a bond pays 90 in year 1 and 10 in year 5. Find each duration.

Q2. Macaulay duration of a project paying 100 in year 1 and 100 in year 2. Spot rates: 1-year 4%, 2-year 5%.

Q3. A basket has 20 kg rice, 10 litres milk, 5 litres fuel. Base-year prices 50, 40, 80; current prices 60, 44, 96. Find the CPI, the inflation rate and the rice weight.

Q4. Base prices: laptop 20, car 50. Last year: 30 laptops, 10 cars. This year: prices 25 and 60, output 40 laptops and 12 cars. Find nominal GDP (both years), real GDP this year (last year's prices), the GDP deflator and real growth.

Q5. Nominal interest 12%, expected inflation 5%. Find the real rate exactly and approximately.

Q6. A bond pays nominal 300% when inflation is 100%. What is the real interest rate? Why does the approximation $i - \pi$ fail here?

Q7. A project pays 1,100 next year. Nominal rate 10%, inflation 4%. Find its present value two ways (nominal, then real). What happens if you wrongly discount 1,100 at the real rate?

Q8. The 3-year zero bond yields 0.97% per year. How much does 50,000 become after 3 years and what is the holding return?

Q9. Estimate the 15-year yield from the 10-year (1.52%) and 20-year (1.94%) yields.

Q10. 1-year rate 3%, 2-year 4%, 3-year 5%. Find the forward rates $r_{1,2}$ and $r_{2,3}$.

Q11. Using 1-year 3% and 2-year 4%: price of a 2-year zero (pays 100) today, its price after one year, and the 1-year return.

Q12. A 20-year zero pays 1000. At 2%, find its price. Find the % change when yield rises 10bp and falls 10bp. Compare with a 1-year zero.

Q13. Name the yield-curve shape: (a) 1-year 5%, 10-year 3%. (b) 1-year 2%, 5-year 3%, 30-year 2%. (c) 1-year 1%, 30-year 4%.

Q14. 1-year rate 5%, 2-year rate 10%. The market expects next year's 1-year rate to be 8%. How much extra over two years does the 2-year bond pay versus rolling 1-year bonds? What does it represent?

Answer key

Q1. (a) (50×1 + 50×10) ÷ 100 = **5.5 years**. (b) (90×1 + 10×5) ÷ 100 = **1.4 years**.

Q2. PV_1 = 100 ÷ 1.04 = 96.15. PV_2 = 100 ÷ 1.05^2 = 90.70. Macaulay = (96.15×1 + 90.70×2) ÷ 186.85 = **1.485 years**.

Q3. Base cost = 1000 + 400 + 400 = 1800. Current cost = 1200 + 440 + 480 = 2120. CPI = **117.78**; inflation **17.78%**; rice weight = 1000 ÷ 1800 = **55.6%**.

Q4. Nominal last year (base prices) = 30×20 + 10×50 = **1,100**. Nominal this year = 40×25 + 12×60 = **1,720**. Real this year = 40×20 + 12×50 = **1,400**. Deflator = 1,720 ÷ 1,400 = **1.229** (prices up about 22.9%). Real growth = 1,400 ÷ 1,100 − 1 = **27.3%** (nominal growth 56.4%).

Q5. Exact: 1.12 ÷ 1.05 − 1 = **6.67%**. Approx: 12 − 5 = **7%**.

Q6. Real = (1 + 3) ÷ (1 + 1) − 1 = **100%**. The shortcut gives 300 − 100 = 200%, wrong, because rates are so large that the cross term matters.

Q7. Nominal: 1,100 ÷ 1.10 = **1,000**. Real: real cash flow = 1,100 ÷ 1.04 = 1,057.69; real rate = 1.10 ÷ 1.04 − 1 = 5.77%; PV = 1,057.69 ÷ 1.0577 = **1,000**. Wrong mix: 1,100 ÷ 1.0577 ≈ **1,040**, which overstates the value.

Q8. 50,000 × 1.0097^3 = **51,469.16**. Holding return = 1.0097^3 − 1 = **2.94%**.

Q9. (1.52 + 1.94) ÷ 2 = **1.73%** (approximate).

Q10. $r_{1,2}$ = 1.04^2 ÷ 1.03 − 1 = **5.01%**. $r_{2,3}$ = 1.05^3 ÷ 1.04^2 − 1 = **7.03%**.

Q11. Today: 100 ÷ 1.04^2 = **92.46**. After one year: 100 ÷ 1.0501 = **95.23**. Return = 95.23 ÷ 92.46 − 1 = **3.00%**, equal to the 1-year rate.

Q12. Price = 1000 ÷ 1.02^{20} = **672.97**. At 2.1%: 659.91 (**−1.94%**). At 1.9%: 686.30 (**+1.98%**). 1-year zero: about −0.098% and +0.098%. The 20-year bond is about 20 times more sensitive.

Q13. (a) **Downward-sloping (inverted)**. (b) **Humped**. (c) **Upward-sloping**.

Q14. Rolling: 1.05 × 1.08 = 1.134 (13.4%). 2-year bond: 1.10^2 = 1.21 (21%). Extra = 1.21 ÷ 1.134 − 1 ≈ **6.7%** over two years. This is the **term premium**, compensation for holding the riskier long-term bond.

Source: lecture slides CF_4 (Time-Varying Rates of Return; reference: Chapter 5 of Ivo Welch, Corporate Finance). Lecture examples use the slide numbers; practice problems are new.